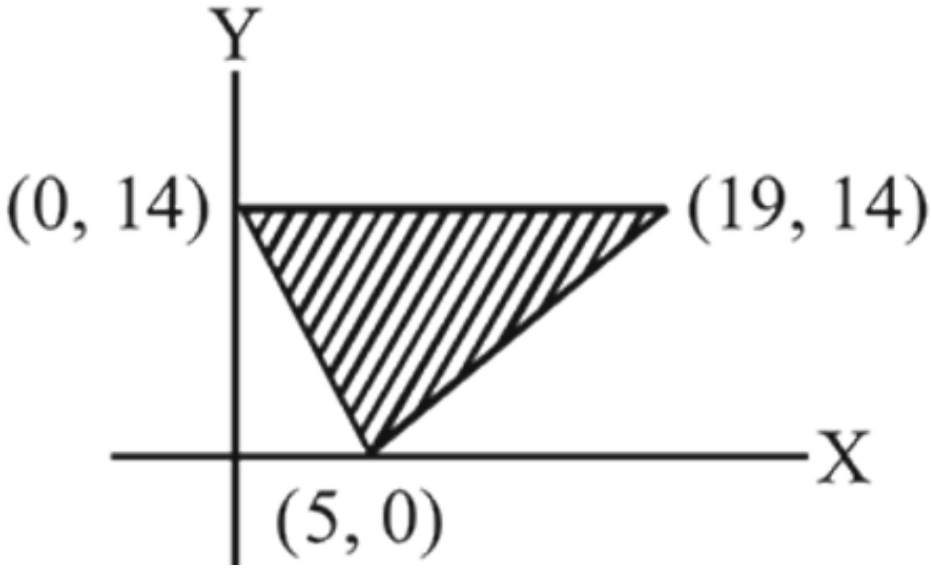
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The shaded region shown in the figure is given by the inequations



(A) $14x + 5y \geq 70, y \leq 14$ and $x - y \geq 5$

(B) $14x + 5y \leq 70, y \leq 14$ and $x - y \geq 5$

(C) $14x + 5y \geq 70, y \geq 14$ and $x - y \geq 5$

(D) $14x + 5y \geq 70, y \leq 14$ and $x - y \leq 5$

Q.2

The number of real values of parameter $\mathbf{k}$ for which $(\log_{16} x)^2 - \log_{16} x + \log_{16} k = 0$ will have exactly one solution is

(A) 0

(B) 2

(C) 1

(D) 4

Q.3

Solve for x, $\frac{|x+3|+x}{x+2} > 1$

(A) $x \in (-5, -2) \cup (-1, \infty)$

(B) $x \in (5, 2) \cup (-1, \infty)$

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(C) $x \in (5, 2)$

(D) $x \in (-1, \infty)$

Q.4

The solution set of the inequality
$37 - (3x + 5) \geq 9x - 8(x - 3)$ is

(A) $(-\infty, 2)$

(B) $(-\infty, -2)$

(C) $(-\infty, 2]$

(D) $(-\infty, -2]$

Q.5

If $\log_{1/2} \frac{x^2+6x+9}{2(x+1)} < -\log_2(x + 1)$, then x lies in the interval

(A) $(-1, 1 + 2\sqrt{2})$

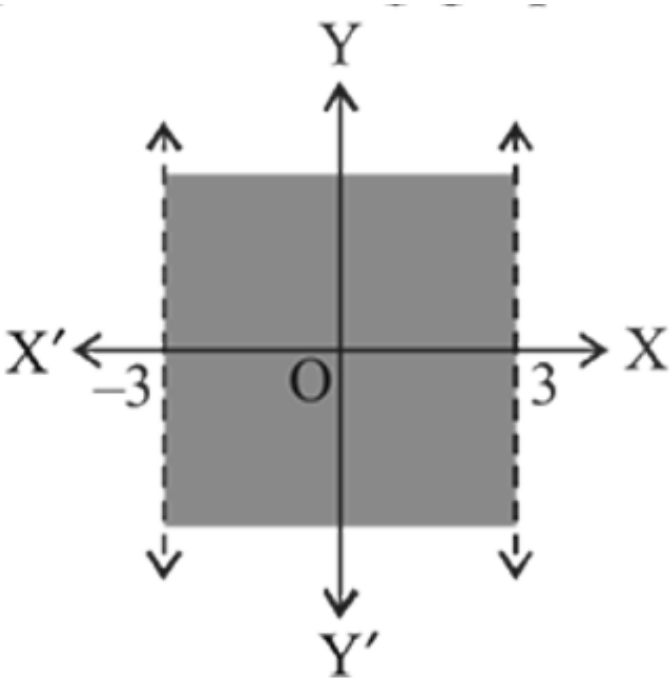
(B) $(1 - 2\sqrt{2}, 2)$

(C) $(-1, \infty)$

(D) None of these

Q.6

The inequality representing the following graph is



(A) $|x| < 3$

(B) $|x| \leq 3$

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(C) $|x| > 3$

(D) $|x| \geq 3$

Q.7

If $5\{x\} = x + [x]$ and $[x] - \{x\} = \frac{1}{2}$ when $\{x\}$ and $[\mathrm{x}]$ are fractional and integral part of x then x is

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) $\frac{5}{2}$

(D) $\frac{7}{2}$

Q.8

Ravi obtained 70 and 75 marks in first two unit tests. Then, the minimum marks he should get in the third test to have an average of at least 60 marks, are

(A) 45

(B) 35

(C) 25

(D) None of these

Q.9

If $\mathrm{R} \geq \mathrm{r} > 0$ and $\mathrm{d} > 0$, then $0 < \frac{\mathrm{d}^2+\mathrm{R}^2-\mathrm{r}^2}{2\mathrm{dR}} \leq 1$

(A) is satisfied if $|\mathrm{d} - \mathrm{R}| \leq \mathrm{r}$

(B) is satisfied if $|\mathrm{d} - \mathrm{R}| \leq 2\mathrm{r}$

(C) is satisfied if $|\mathrm{d} - \mathrm{R}| \geq \mathrm{r}$

(D) is not satisfied at all

Q.10

The length of a rectangle is three times the breadth. If the minimum perimeter of the rectangle is 160 cm, then what can you say about breadth?

(A) breadth $= 20$

(B) breadth ≤ 20

(C) breadth ≥ 20

(D) breadth $\neq 20$

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Q.11

$3^{-\log_{1/3}(4)} \quad (0.1)^{\log_{0.01}(4)} \quad 7^{\log_7(3)}$ is equal to

(A) 0

(B) $5^{\sqrt{\log_5 3}}$

(C) $2.5^{\sqrt{\log_5 3}}$

(D) None of these

Q.12

The solution set of $(x)^2 + (x+1)^2 = 25$, where (x) is the least integer greater than or equal to x, is

(A) $(2, 4)$

(B) $(-5, -4] \cup (2, 3]$

(C) $[-4, -3) \cup [3, 4)$

(D) None of these

Q.13

Solution of $|x-1| \geq |x-3|$ is

(A) $x \leq 2$

(B) $x \geq 2$

(C) $[1, 3]$

(D) None of these

Q.14

The solution set of $\frac{2x-1}{3} \geq \left(\frac{3x-2}{4}\right) - \left(\frac{2-x}{5}\right)$ is $(-\infty, a]$. The value of 'a' is

(A) 2

(B) 3

(C) 4

(D) 5

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Q.15

For positive real numbers a, b, c such that $a + b + c = p$, which one does not hold?

(A) $(p-a)(p-b)(p-c) \leq \frac{8}{27}p^3$

(B) $(p-a)(p-b)(p-c) \geq 8abc$

(C) $\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \leq p$

(D) None of these

Q.16

If x satisfies the inequalities $x + 7 < 2x + 3$ and $2x + 4 < 5x + 3$, then x lies in the interval

(A) $(-\infty, 3)$

(B) $(1, 3)$

(C) $(4, \infty)$

(D) $(-\infty, -1)$

Q.17

The solution set of the inequality $5^{x+2} > \left(\frac{1}{25}\right)^{1/x}$ is

(A) $(-2, 0)$

(B) $(-2, 2)$

(C) $(-5, 5)$

(D) $(0, \infty)$

Q.18

The solution set of the inequality $|x + 2| - |x - 1| < x - \frac{3}{2}$ is

(A) $\left(\frac{9}{2}, \infty\right)$

(B) $\left(-\infty, \frac{3}{2}\right)$

(C) $\left(-2, -\frac{3}{2}\right)$

(D) $\left(-1, \frac{3}{2}\right)$

Q.19

$\frac{3x-4}{2} \geq \frac{x+1}{4} - 1$, then $x \in$

(A) $[1, \infty)$

(B) $(1, \infty)$

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(C) $(-5, 5)$

(D) $[-5, 5]$

Q.20

Solution set of the inequality $\log_3(x+2)(x+4) + \log_{1/3}(x+2) < \frac{1}{2}\log_{\sqrt{3}} 7$ is

(A) $(-2, -1)$

(B) $(-2, 3)$

(C) $(-1, 3)$

(D) $(3, \infty)$

Answers & Solutions

Q.1 Answer:

$14x + 5y \geq 70, y \leq 14$ and $x - y \leq 5$

Solution:

Consider the equation of the line joining $(0, 14)$ and $(5, 0)$. Applying intercept form gives us

$\frac{x}{5} + \frac{y}{14} = 1$ or $14x + 5y = 70$

Now it has been shaded away from the origin, hence the inequality has to be false at $\mathrm{x, y} = (0, 0)$.

At $\mathrm{x, y} = (0, 0)$ **we get** $14\mathrm{x} + 5\mathrm{y} = 0$. **Therefore for false inequality,** $14\mathrm{x} + 5\mathrm{y} \geq 70$

Similarly the equation of the line joining $(0, 14)$ and $(19, 14)$, is given by $y = 14$.

Since it is shaded towards the origin, hence $y \leq 14$.

The equation of the line joining $(5, 0)$ and $(19, 14)$ is given by

$\frac{\mathrm{y}-14}{\mathrm{x}-19} = \frac{\mathrm{y}-0}{\mathrm{x}-5}$

$\mathrm{xy} - 14\mathrm{x} - 5\mathrm{y} + 70 = \mathrm{xy} - 19\mathrm{y}$

$14\mathrm{x} - 14\mathrm{y} = 70$ **or** $\mathrm{x} - \mathrm{y} = 5$. **Since it is shaded towards the origin,** $\mathrm{x} - \mathrm{y} \leq 5$

Hence the shaded area is given by $14\mathrm{x} + 5\mathrm{y} \geq 70, \mathrm{y} \leq 14, \mathrm{x} - \mathrm{y} \leq 5$

Q.2 Answer:

1

Solution:

Let $\log_{16} \mathrm{x} = \mathrm{t}$

Hence, $t^2 - t + \log_{16} k = 0$ **and change** $k > 0$

The equation has exactly one solution. Hence, the discriminant must be zero.

$\Delta = 1 - 4\log_{16} \mathrm{k}$

$\Rightarrow 0 = 1 - 4\log_{16} \mathrm{k}$

$\Rightarrow \log_{16} \mathrm{k} = \frac{1}{4}$

$\Rightarrow \mathrm{k} = 2$

Q.3 Answer:

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$x \in (-5,-2) \cup (-1,\infty)$

Solution:

We have $\frac{|x+3|+x}{x+2} > 1 \Rightarrow \frac{|x+3+x|}{x+2} - 1 > 0$

$\Rightarrow \frac{|x+3|-2}{x+2} > 0$

Now two cases arise :

Case I : When $x+3 \geq 0$, i.e., $x \geq -3$. Then $\frac{|x+3|-2}{x+2} > 0 \Rightarrow \frac{x+3-2}{x+2} > 0$

$\Rightarrow \frac{x+1}{x+2} > 0 \Rightarrow \{(x+1) > 0$ and $x+2 > 0\}$

or $\{x+1 < 0$ and $x+2 < 0\}$

$\Rightarrow \quad \{x > -1$ and $x > -2\}$ or $\{x < -1$ and $x < -2\}$

$\Rightarrow \quad x > -1$ or $x < -2 \Rightarrow x \in (-1,\infty)$

or $x \in (-\infty,-2)$

$\Rightarrow \quad x \in (-3,-2) \cup (-1,\infty)$ [Since $x \geq -3] \ldots (1)$

Case II : When $x+3 < 0$, i.e., $x < -3$ $\frac{|x+3|-2}{x+2} > 0 \Rightarrow \frac{-x-3-2}{x+2} > 0$

$\Rightarrow \quad \frac{-(x+5)}{x+2} > 0 \Rightarrow \frac{x+5}{x+2} < 0$

$\Rightarrow (x+5 < 0$ and $x+2 > 0)$

or $\quad (x+5 > 0$ and $x+2 < 0)$

$\Rightarrow \quad (x < -5$ and $x > -2)$

or $(x > -5$ and $x < -2$). It is not possible. $\Rightarrow \quad x \in (-5,-2)$ Combining (1) and (2), the required solution is $x \in (-5,-2) \cup (-1,\infty)$

Q.4 Answer:

$(-\infty, 2]$

Solution:

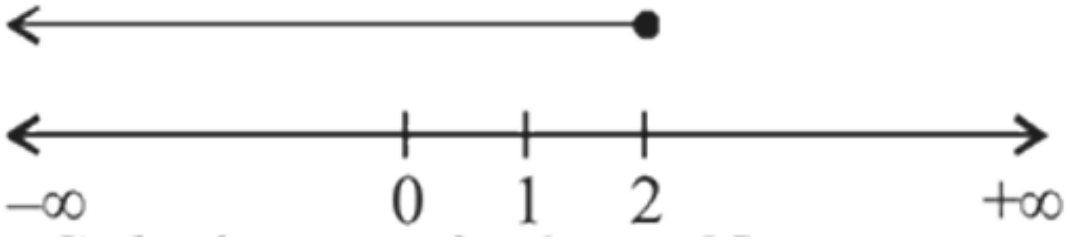
We have, $37-(3x+5) \geq 9x-8(x-3)\ (37-3x-5) \geq 9x-8x+24$

$\Rightarrow 32-3x \geq x+24$

Transferring the term 24 to L.H.S. and the term $(-3x)$ to R.H.S. $32-24 \geq x+3x \Rightarrow 8 \geq 4x \Rightarrow 4x \leq 8$

Dividing both sides by 4 ,

$\frac{4x}{4} \leq \frac{8}{4} \Rightarrow x \leq 2$



$\therefore$ Solution set is $(-\infty, 2]$.

Q.5 Answer:

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$(-1, 1+2\sqrt{2})$

Solution:

The log functions are defined if $\frac{x^2+6x+9}{2(x+1)} > 0$ **and** $x+1>0 \Rightarrow \frac{(x+3)^2}{2(x+1)} > 0$ **and** $x+1>0 \Rightarrow$
$x > -1$

Now the inequality is $\log_{2^{-1}} \frac{x^2+6x+9}{2(x+1)} < -\log_2(x+1)$

$\Rightarrow -\log_2 \frac{x^2+6x+9}{2(x+1)} < \log_2(x+1)$

$\Rightarrow \log_2 \frac{x^2+6x+9}{2(x+1)} > \log_2(x+1) \Rightarrow \frac{x^2+6x+9}{2(x+1)} > (x+1)$

$\Rightarrow \frac{-x^2+2x+7}{2(x+1)} > 0 \Rightarrow (x+1)\left(x^2-2x-7\right) < 0$

$\Rightarrow x^2-2x-7 < 0$

$\Rightarrow 1-2\sqrt{2} < x < 1+2\sqrt{2}$

but $x > -1 \Rightarrow -1 < x < 1+2\sqrt{2}$

Q.6 Answer:

$|x| < 3$

Solution:

The shaded region in the figure lies between $x=-3$ **and** $x=3$ **not including the line** $x=-3$ **and** $\mathrm{x}=3$ **(lines are dotted). Therefore,** $-3 < \mathrm{x} < 3 \Rightarrow \quad |x| < 3 \quad [\because |x| < a \Leftrightarrow -a < x < a]$

Q.7 Answer:

$\frac{3}{2}$

Solution:

$5\{x\} = x + [x]$ **and** $[x] - \{x\} = \frac{1}{2}$

since $x = [x] + \{x\} \Rightarrow 4\{x\} = 2[x]$ **and**

$[\mathrm{x}] - \{\mathrm{x}\} = \frac{1}{2}$

after solving $[\mathrm{x}] = 1$ **and** $\{\mathrm{x}\} = \frac{1}{2} \therefore \mathrm{x} = 1 + \frac{1}{2} = \frac{3}{2}$.

Q.8 Answer:

35

Solution:

Let Ravi got x **marks in third unit test.** $\therefore$ **Average marks obtained by Ravi** $= \frac{\text{Sum of marks in all tests}}{\text{Number of tests}}$
$= \frac{70+75+x}{3} = \frac{145+x}{3}$

Now, it is given that he wants to obtain an average of at least 60 marks. At least 60 marks means that the marks should be greater than or equal to 60**. i.e.** $\frac{145+\mathrm{x}}{3} \geq 60$

$\Rightarrow 145 + \mathrm{x} \geq 60 \times 3$

$\Rightarrow 145 + x \geq 180$

Now, transferring the term 145 to R.H.S., $x \geq 180 - 145 \Rightarrow x \geq 35$

i.e. Ravi should get greater than or equal to 35 marks in third unit test to get an average of at least 60

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marks.

$\therefore$ **Minimum marks Ravi should get** $= 35.$

Q.9 Answer:

is satisfied if $|d - R| \leq r$

Solution:

Given $R \geq r > 0$ **and** $d > 0 \Rightarrow \quad 0 < \frac{d^2+R^2-r^2}{2dR} \leq 1$

$\Rightarrow \quad 0 < (d + R - r)(d + R + r) \leq 2dR$; **which is true if**

$(d^2 + R^2 - r^2) \leq 2dR$, **which is true iff** $d^2 + R^2$

$-2dR \leq r^2$

$\Rightarrow (d - R)^2 \leq r^2$

$|d - R| \leq r$, **which is also** $-r \leq (d - R) \leq r$

$|d - R| \leq r$, **which is also** $- r \leq (d - R) \leq r$

Q.10 Answer:

breadth ≥ 20

Solution:

If x **cm is the breadth, then** $2(3x + x) \geq 160 \Rightarrow x \geq 20$

Q.11 Answer:

0

Solution:

Let

$$D = \begin{matrix} 5^{\sqrt{\log_5 3}} & 5^{\sqrt{\log_5 3}} & 5^{\sqrt{\log_5 3}} \\ 3^{-\log_{1/3}(4)} & (0.1)^{\log_{0.01}(4)} & 7^{\log_7(3)} \\ 7 & 3 & 5 \end{matrix}$$

$C_2 \leftrightarrow C_2 - C_1$ **and** $C_3 \leftrightarrow C_3 - C_1$

$$\Rightarrow D = \begin{matrix} 5^{\sqrt{\log_5 3}} & 5^{\sqrt{\log_5 3}} - 5^{\sqrt{\log_5 3}} & 5^{\sqrt{\log_5 3}} - 5^{\sqrt{\log_5 3}} \\ 3^{-\log_{1/3}(4)} & (0.1)^{\log_{0.01}(4)} - 3^{-\log_{1/3}(4)} & 7^{\log_7(3)} - 3^{-\log_{1/3}(4)} \\ 7 & -4 & -2 \end{matrix}$$

$$\Rightarrow D = \begin{matrix} 5^{\sqrt{\log_5 3}} & 0 & 0 \\ 3^{-\log_{1/3}(4)} & (0.1)^{\log_{0.01}(4)} - 3^{-\log_{1/3}(4)} & 7^{\log_7(3)} - 3^{-\log_{1/3}(4)} \\ 7 & -4 & -2 \end{matrix}$$

$$\Rightarrow D = \begin{matrix} 5^{\sqrt{\log_5 3}} & 0 & 0 \\ 3^{-\log_{1/3}(4)} & 4^{\log_{0.01}0.1} - 4^{-\log_{1/3}3} & 3^{\log_7 7} - 4^{-\log_{1/3}3} \\ 7 & -4 & -2 \end{matrix}$$

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$\because c^{\log_a b} = b^{\log_a c}$

$$\Rightarrow D = \begin{matrix} 5\sqrt{\log_5 3} & 0 & 0 \\ 3^{-\log_{1/3}(4)} & 4^{\frac{1}{2}} - 4^1 & 3^1 - 4^1 \\ 7 & -4 & -2 \end{matrix}$$

$\because \log_a b = \frac{1}{\log_b a}$

$$\Rightarrow D \begin{matrix} 5\sqrt{\log_5 3} & 0 & 0 \\ 4 & -2 & -1 \\ 7 & -4 & -2 \end{matrix} = 0$$

Q.12 Answer:

$(-5, -4] \cup (2, 3]$

Solution:

$(\mathrm{x})^2 + (\mathrm{x}+1)^2 = 25 \Rightarrow (\mathrm{x})^2 + \{(\mathrm{x})+1\}^2 = 25$
$\Rightarrow 2(\mathrm{x})^2 + 2(\mathrm{x}) - 24 = 0$
$\Rightarrow (x)^2 + (x) - 12 = 0 \Rightarrow (x) = -4$ **or** 3
$\mathrm{Now}(x) = -4 \Rightarrow -5 < x \leq -4$ **and**
$(x) = 3 \Rightarrow 2 < x \leq 3$
$\therefore$ **Solution set is** $(-5, -4] \cup (2, 3]$

Q.13 Answer:

$x \geq 2$

Solution:

$|x-1|$ **is the distance of** x **from 1 .** $|x-3|$ **is the distance of** x **from 3 The point** $\mathrm{x} = 2$ **is equidistant from 1 and 3 . Hence, the solution consists of all** $\mathrm{x} \geq 2$**.**

Q.14 Answer:

2

Solution:

$\frac{2x-1}{3} \geq \frac{15x-10-8+4x}{20}$
$\Rightarrow \quad \frac{2x-1}{3} \geq \frac{19x-18}{20}$
$\Rightarrow \quad 40x - 20 \geq 57x - 54 \Rightarrow -17x \geq -34 \Rightarrow x \leq 2$
$\Rightarrow \quad x \in (-\infty, 2]$

Q.15 Answer:

$\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \leq p$

Solution:

Using A.M.\geq G.M. one can show that $(b+c)(c+a)(a+b) \geq 8abc$
$\Rightarrow \quad (p-a)(p-b)(p-c) \geq 8abc$

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holds. Also, Therefore, $\frac{(p-a)+(p-b)+(p-c)}{3} \geq [(p-a)$
$(p-b)(p-c)]^{1/2}$
or $\frac{3p-(a+b+c)}{3} \geq [(p-a)(p-b)(p-c)]^{1/3}$
or $\frac{2p}{3} \geq [(p-a)(p-b)(p-c)]^{1/3}$
or $(p-a)(p-b)(p-c) \leq \frac{8p^3}{27}$
Therefore, holds. Again,
$\frac{1}{2}\left(\frac{bc}{a}+\frac{ca}{b}\right) \geq \sqrt{\left(\frac{bc}{a}\frac{ca}{b}\right)}$
and so on. Adding the inequalities, we get
$\frac{bc}{a}+\frac{ca}{b}+\frac{ab}{c} \geq a+b+c=p$
Therefore, does not hold.

Q.16 Answer:

$(4, \infty)$

Solution:

$x+7<2x+3$
$2x+4<5x+3$
$\Rightarrow 4<\mathrm{x} \quad 1<3\mathrm{x} \Rightarrow \mathrm{x}>\frac{1}{3}$
so, $x>\frac{1}{3}$ and $x>4 \therefore$ x lies in $(4, \infty)$

Q.17 Answer:

$(0, \infty)$

Solution:

We have
$5^{x+2}>\left(\frac{1}{25}\right)^{\frac{1}{x}} \Rightarrow 5^{x+2}>5^{-\frac{2}{x}} \Rightarrow x+2>-\frac{2}{x}$
$\left[\because \text{base } 5>1.\right.$
$\Rightarrow x+2+\frac{2}{x}>0 \Rightarrow \frac{x^2+2x+2}{x}>0 \Rightarrow \frac{1}{x}>0$
$\left[\because \mathrm{x}^2+2\mathrm{x}+2>0 \forall \mathrm{x} \in \mathrm{R}\right]$
$\Rightarrow \mathrm{x}>0$ or $\mathrm{x} \in (0, \infty)$

Q.18 Answer:

$\left(\frac{9}{2}, \infty\right)$

Solution:

The inequality is $|x+2|-|x-1|<x-\frac{3}{2}$ Dividing the problem into three intervals :
(i) If $x<-2$, then $-(x+2)+(x-1)<x-\frac{3}{2}$
$\Rightarrow x>-\frac{3}{2}$
But $-\frac{3}{2}>-2$, hence no common values
$\Rightarrow \mathrm{x} \in \phi$
(ii) If $-2 \leq x<1$, then $(x+2)+(x-1)<x-\frac{3}{2} \Rightarrow x<-\frac{5}{2}$

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But $-\frac{5}{2} < -2$, hence no common values $\Rightarrow x \in \phi$

(iii) If $x \geq 1$, then $(x+2)-(x-1) < x - \frac{3}{2}$

$\Rightarrow x > \frac{9}{2}$

$\because \frac{9}{2} > 1$

$\Rightarrow$ common solution is $x > \frac{9}{2} \Rightarrow x \in \left(\frac{9}{2}, \infty\right)$

$\therefore$ Solution set is $x \in \left(\frac{9}{2}, \infty\right)$

Q.19 Answer:

$[1, \infty)$

Solution:

We have $\frac{3x-4}{2} \geq \frac{x+1}{4} - 1$

or $\frac{3x-4}{2} \geq \frac{x-3}{4}$

or $2(3x-4) \geq (x-3)$ or $6x - 8 \geq x - 3$

or $5x \geq 5$ or $x \geq 1$

Thus, all real numbers which are greater than or equal to 1 is the solution set of the given inequality.

$\therefore x \in [1, \infty)$

Q.20 Answer:

$(-2, 3)$

Solution:

$\log_3(x+2)(x+4) + \log_{1/3}(x+2) < (1/2)\log_{\sqrt{3}} 7$

Clearly, this is defined for $x > -2 \Rightarrow$ (i)

$\Rightarrow \frac{\log(x+2)}{\log 3} + \frac{\log(x+4)}{\log 3} + \frac{\log(x+2)}{\log\left(\frac{1}{3}\right)} < \frac{1}{2}\frac{\log 7}{\frac{1}{2}\log 3}, \left[\because \log(ab) = \log a + \log b \& \log_a b = \frac{1}{\log_b a}\right]$

$\Rightarrow \frac{\log(x+2)}{\log 3} + \frac{\log(x+4)}{\log 3} - \frac{\log(x+2)}{\log 3} < \frac{\log 7}{\log 3}$

$\Rightarrow \frac{\log(x+4)}{\log 3} < \frac{\log 7}{\log 3}$

$\Rightarrow \log(x+4) < \log(7), \because \log 3 > 0$

$\Rightarrow x + 4 < 7$

$\Rightarrow x < 3 \Rightarrow$ **(ii)**

Thus from (i) and (ii)

The solution set is $(-2, 3)$

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